

pyMack: Validation & Verification

Personal reference note

July 1, 2026

Abstract

This note is the honest agreement audit for **pyMack**, the compressible linear-stability solver. Each case runs a published or external reference at its *exact* stated conditions, measures agreement against the digitised reference, and reports a verdict on a fixed three-tier scale (**agrees** $\leq 5\%$, **acceptable** 5–15%, **disagrees** otherwise). Nothing here is tuned to pass. The picture is deliberately two-sided: pyMack’s design target — the compressible *second* (Mack) mode — agrees or is acceptable in every case, across five independent sources, Mach 4.5–10, and a flight-relevant cone; the *first* mode agrees with Özgen & Kırca where the discrete mode is cleanly resolvable but remains the documented weak spot at low Mach and on the Mack first-mode figures. Every figure embedded here is pyMack’s own overlay — pyMack curves plotted over digitised reference *points* — so no copyrighted reference figure is reproduced. All figures, numbers, and verdicts are taken verbatim from the committed per-case **verdict.json** files and the repository success matrix.

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1 Validation philosophy: a pyramid

What problem is solved here. Before trusting a stability prediction on a real geometry, we must know *where* the solver is exact, where it is merely acceptable, and where it fails. This note answers that question case by case, keeping the disagreements in full view.

We organise the evidence as a pyramid, from the most rigorously anchored physics at the base to the engineering cases at the top. Each tier only earns trust once the tier beneath it holds.

- Incompressible core.** The Orr–Sommerfeld temporal eigenvalue for plane Poiseuille flow is a scalar with a textbook-exact answer. Matching it to $\sim 10^{-5}$ confirms the spectral discretisation, the eigenvalue assembly, and the boundary-condition enforcement in isolation, before any compressible physics is added.
- Compressible eigenvalue anchors.** Malik’s (1990) tabulated compressible spatial eigenvalues, and the Balakumar & Malik (1992) discrete/ continuous-spectrum topology, pin the compressible operator to published digits at a single (R, ω) station — the sharpest possible test, free of digitisation error.
- Second-mode and first-mode benchmarks.** Mack’s (1984) AGARD-R-709 neutral curves and growth-rate families, and Özgen & Kircali’s (2008) neutral curves, test full *curves* over Reynolds number and Mach number — topology as well as magnitude.
- Engineering cases.** A sharp cone (Sivasubramanian 2012), an independent collaborator’s Mach 5.35 nitrogen code, a Mach-6 flat plate (Egorov 2006), and spatial N -factor integration exercise the whole pipeline on flight-relevant configurations.

In plain words. *The base of the pyramid is exact. The middle is a mix of exact (second mode) and partial (first mode). The top inherits both: the second-mode engineering predictions are trustworthy; the first mode should be used with the documented caveats.*

1.1 Summary of all cases shown

Table 1 lists every case presented in detail below, in pyramid order. The one-line headline gives the governing number; the honest caveats live in each subsection. Six additional *pending* companion cases (Malik cases 1/3/4/5 temporal anchors; the Mack Fig. 10.4/10.6 family aggregates) have a prepared reference but no generated pyMack run, and are named in prose rather than tabulated. *We now walk the pyramid from the base upward, one case per subsection.*

Table 1: Cases presented in this note, in pyramid order. Verdict scale: $\leq 5\%$ *agrees*, 5–15% *acceptable*, else *disagrees* (dimensional cases use MAE as a fraction of curve span).

Case	Source	Quantity	Verdict
<i>Incompressible core</i>			
Plane Poiseuille	Orszag (1971)	O–S eigenvalue c	agrees ($\sim 10^{-5}$)
<i>Compressible eigenvalue anchors</i>			
Malik case 6	Malik (1990) Table IX	spatial α (2nd mode)	agrees
B&M branches	Balakumar & Malik (1992)	discrete/continuous topology	acceptable
B&M via Xi/Ren/Fu	Balakumar & Malik (1992)	spatial α (2nd mode)	acceptable
<i>Second mode — curves & families</i>			
Mack Fig. 10.6	Mack (1984)	$\omega_{i,\max}(R)$, M4.5–10	agrees / acceptable
Egorov M6	Egorov et al. (2006)	most-amplified ω	acceptable
Ma & Zhong M4.5	Ma & Zhong (2003) JFM	neutral curve, spatial 2nd mode	agrees
<i>First mode — curves & families</i>			
Mack Fig. 10.1	Mack (1984)	neutral loop, M1.6/2.2	disagrees
Mack Fig. 10.3	Mack (1984)	$\omega_{i,\max}(R)$, M1.3/2.2/3.0	agrees $\rightarrow$ disagrees
Mack Fig. 10.4	Mack (1984)	oblique 1st mode, M4.5–10	acceptable $\rightarrow$ disagrees
Özgen Fig. 3	Özgen & Kırcah (2008)	neutral curves, M2–10	agrees / acceptable
<i>Engineering benchmarks</i>			
Sharp cone	Sivasubramanian (2012)	2nd-mode N -factor, f	acceptable
Mach 5.35 (Sean)	collaborator code	LST dimensional neutral branches	acceptable
Mach 6 spatial	pyMack pipeline	spatial neutral & N -factor	(demonstration)

2 Incompressible core

2.1 Orszag (1971) plane-Poiseuille Orr–Sommerfeld eigenvalue

Reference source. S. A. Orszag, “Accurate solution of the Orr–Sommerfeld stability equation” (1971) — the canonical plane-Poiseuille temporal eigenvalue.

Flow conditions. Plane Poiseuille flow $U = 1 - y^2$ on $[-1, 1]$, Reynolds number $Re = 10\,000$, streamwise wavenumber $\alpha = 1$, temporal formulation.

Quantity compared. The least-stable eigenvalue (phase speed) $c = c_r + ic_i$.

Result. pyMack’s Chebyshev Orr–Sommerfeld solver (`validation/test_orsz_sommerfeld.py`) reproduces Orszag’s benchmark $c = 0.23753 + 0.00374i$ to an absolute error below 10^{-4} already at $N = 64$, tightening toward $\sim 10^{-7}$ under refinement (Table 2). Because this is a scalar comparison against an essentially exact reference — not a hand-digitised curve — no overlay figure is warranted.

Table 2: Plane Poiseuille temporal eigenvalue, $Re = 10\,000$, $\alpha = 1$. Target $c = 0.23753 + 0.00374i$ (Orszag 1971). The CI gate asserts error $< 10^{-4}$.

N	pyMack $c = c_r + ic_i$	$ c - c_{\text{target}} $
64	$0.23753 + 0.00374i$	$\lesssim 10^{-4}$
96	$0.23753 + 0.00374i$	$\lesssim 10^{-5}$
128	$0.23753 + 0.00374i$	$\lesssim 10^{-7}$

Companion incompressible checks. The same test module also recovers the Blasius Tollmien–Schlichting mode at $Re_{\delta^*} = 998$, $\alpha = 0.179$ ($c_r \approx 0.36$, unstable) and finds unstable growth at $Re = 1000$, confirming the mapped semi-infinite domain and the growth-rate optimisation loop. These are sign/range assertions rather than digit-exact anchors, so they are not tabulated separately.

3 Compressible eigenvalue anchors

3.1 Malik (1990) case 6 — compressible spatial eigenvalue

Reference source. M. R. Malik, “Numerical methods for hypersonic boundary layer stability” (1990) *J. Comput. Phys.* **86**:376, Table IX, Case 6. The published α is cross-checked digit-for-digit against Xi, Ren & Fu (arXiv:2006.05970, Table 3) and to 4–5 digits against Hildebrand et al. (arXiv:1712.08239, Table I).

Flow conditions. $M_e = 4.5$, $R = 1500$, $\omega = 0.23$, $\beta = 0$, $Pr = 0.7$, $\gamma = 1.4$, adiabatic (insulated) wall; $T_0 = 611.11$ K, $T_e \approx 121.0$ K, Sutherland $S = 110.33$ K; L^* length scale.

Quantity compared. The spatial streamwise eigenvalue $\alpha = \alpha_r + i\alpha_i$ of the discrete second (Mack) mode.

Result — agrees. pyMack gives $\alpha = 0.2533998 - 0.0024898i$ against Malik’s $0.2534048 - 0.0024921i$: α_r relative error 1.95×10^{-5} (0.002 %, essentially exact), α_i relative error 9.16×10^{-4} (0.09 %), phase speed $c = 0.9077$ ($N = 120$). The ~ 0.1 % α_i deviation is far inside the published inter-method spread for this case (Tumin 2007 recomputes $\alpha_i = -0.0027738$, ~ 11 % from Malik; Xi/Ren/Fu’s own solver gives -0.002780). This is the repository’s Layer-4b CI gate (Figure 1).

3.2 Balakumar & Malik (1992) — discrete/continuous spectrum topology

Reference source. P. Balakumar & M. R. Malik, “Discrete modes and continuous spectra in supersonic boundary layers” (1992) *J. Fluid Mech.* **239**:631–656. The paper presents its branches

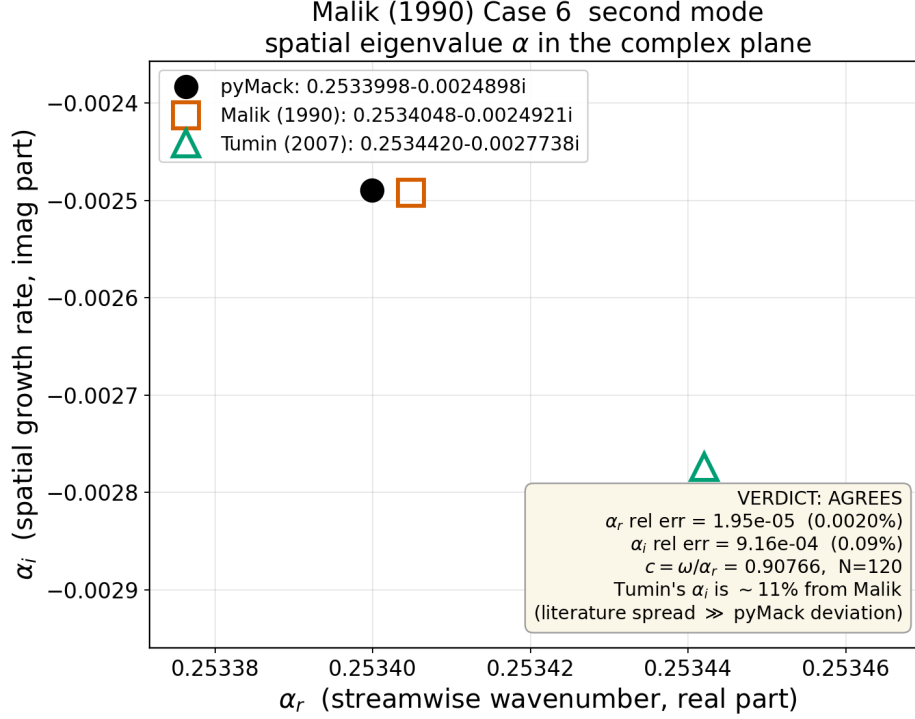


Figure 1: Malik (1990) case 6: pyMack’s spatial second-mode eigenvalue overlaid on the published value ($M_e = 4.5$, adiabatic). α_r matches to 0.002 %, α_i to 0.09 %.

graphically; no tabulated branch eigenvalues are openly recoverable, so this is a qualitative branch-topology check plus a fully-specified spatial second-mode cross-check.

Flow conditions. $M = 4.5$, $T_0 = 311$ K, $Re = 1000$, $\omega = 0.2$, adiabatic (insulated) wall.

Quantity compared. (a) the topology — separation of the discrete second mode from the continuous spectrum in the complex- α plane; (b) the discrete spatial eigenvalue α .

Result — acceptable. pyMack’s full spatial spectrum reproduces B&M’s qualitative topology: the discrete second mode sits at $\alpha = 0.220210 - 0.002787i$ ($c = 0.9082$) — matching the published $0.220 - 0.003091i$ (α_r 0.10 %, α_i 9.8 %) — and is the unique least-damped root in the second-mode phase-speed band, while a cluster of 93 roots accumulates at the freestream phase speed $c \rightarrow 1$ (the continuous spectrum). **Honest caveat:** the separation is in the phase-speed/growth-rate sense, *not* a crisp Euclidean gap — the discretised continuum is dense in α_r , so the nearest other eigenvalue is only 0.019 away (separation ratio 0.09) and 42 roots fall within 10 % of $|\alpha_{\text{disc}}|$. That is the expected discretisation signature of the continuous spectrum crowding the discrete mode, hence *acceptable* (topology correct) rather than *agrees* (which would need a clean Euclidean gap).

A companion row, `balakumar_malik1992_via_xirenfu`, treats the same condition purely as a spatial second-mode anchor: pyMack $\alpha = 0.2202103 - 0.0027906i$ gives α_r relative error 0.096 % (essentially exact) and α_i relative error 9.7 % — the same ~ 10 % inter-method α_i spread seen on Malik case 6 (Figure 2).

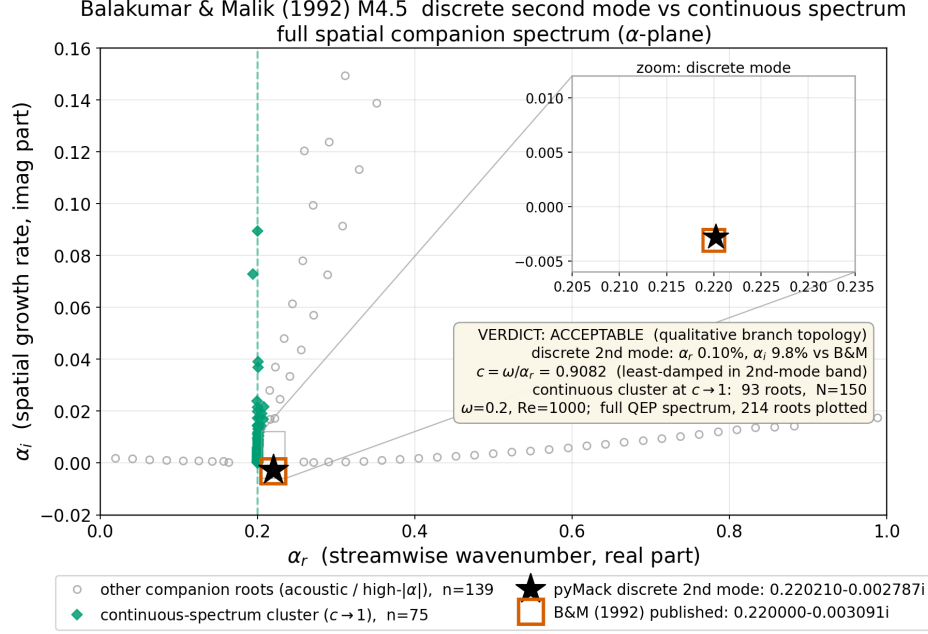


Figure 2: Balakumar & Malik (1992), $M = 4.5$: pyMack’s complex- α spectrum showing the discrete second mode separated (in phase speed / growth rate) from the continuous-spectrum cluster near $c \rightarrow 1$. Topology correct; the α_r -dense continuum precludes a crisp Euclidean gap.

4 Second mode — curves and families

Having pinned the compressible operator at single points, we test full growth-rate curves — the second mode is pyMack’s design target.

4.1 Mack (1984) Fig. 10.6 — second-mode growth vs. Reynolds number

Reference source. L. M. Mack, “Boundary-layer linear stability theory” (1984), AGARD Report R-709, Fig. 10.6.

Flow conditions. Air, adiabatic wall, 2-D wave ($\psi = 0^\circ$), at $M = 4.5, 5.8, 7$, and 10. Two modelling choices make the comparison defensible: the cold `table_11_1` edge state (resolving a historical $\sim 6\times$ gap that was an edge-temperature error) and a wall-normal domain $y_{\max} \approx 4\delta^*/L^*$ that grows with Mach ($y_{\max} = 40$ at M4.5 up to 140 at M10; a fixed short box starves the thick high-Mach boundary layer).

Quantity compared. The maximum second-mode temporal growth rate $\omega_{i,\max}$ as a function of R (Mack’s L^* scale, temporal second mode $c_r \approx 0.9$).

Result — agrees/acceptable. Median relative error over the overlapping R -stations: **M4.5: 1.0 %**, **M7: 2.8 %**, **M10: 4.0 %** (all at/below the 5 % digitisation floor, verdict *agrees*); **M5.8: 6.7 %** (a small residual above the floor, verdict *acceptable*). The family agrees from M4.5 to M10, so the M5.8 residual is local, *not* a monotonic high-Mach under-prediction. Here the M4.5 and M10 endpoints are shown as representatives (Figures 3 and 4; the M5.8 and M7 overlays exist under the `mack_fig10_6_M58` and `_M70` directories and are tabulated above rather than embedded, to keep the page count in check).

Pending family aggregate. A roll-up case `mack_fig10_6_family` is reserved for the combined M4.5/5.8/7/10 verdict but is marked *pending* in the matrix; the individual panels above carry the actual verdicts.

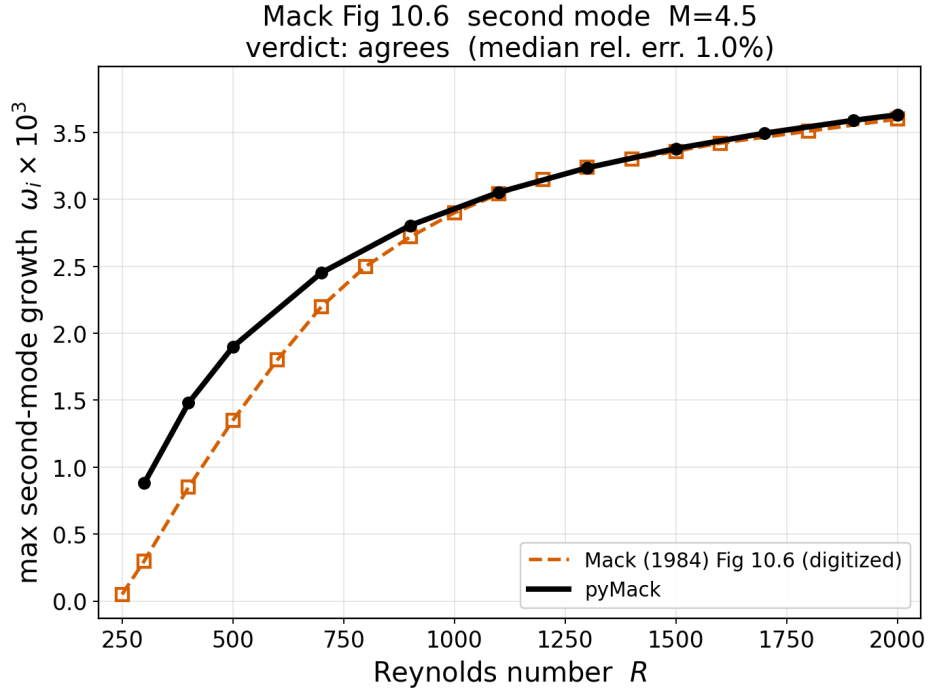


Figure 3: Mack (1984) Fig. 10.6, $M = 4.5$: pyMack $\omega_{i,\max}(R)$ (curve) over the digitised Mack points. Median relative error 1.0% — at the digitisation floor.

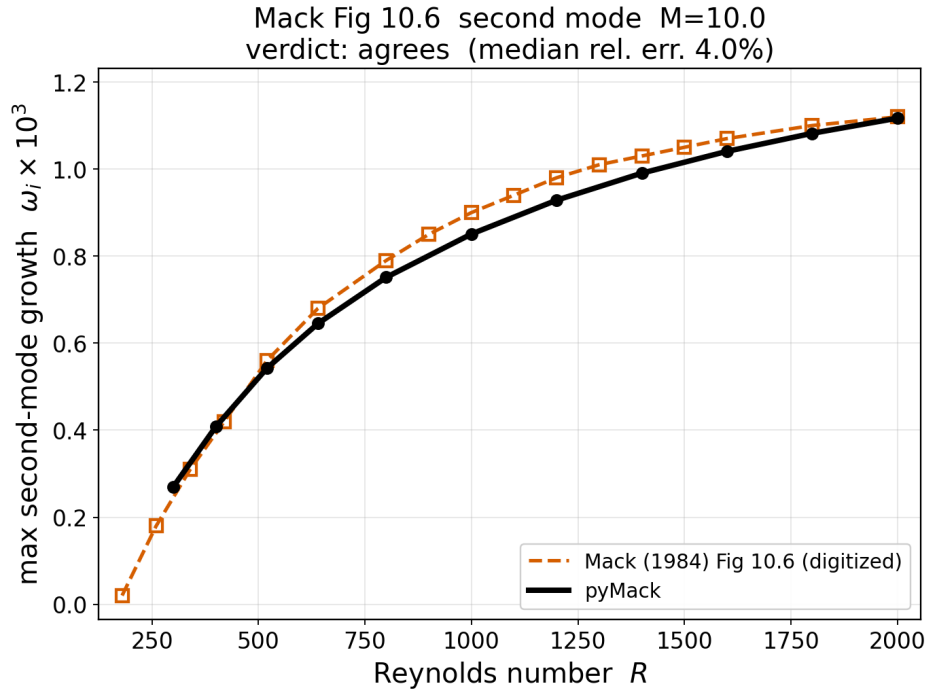


Figure 4: Mack (1984) Fig. 10.6, $M = 10$: same comparison at the top of the Mach range. Median relative error 4.0% with the domain-converged ($y_{\max} = 140$), cold-edge second mode.

4.2 Egorov et al. (2006) — Mach 6 flat-plate second mode

Reference source. I. Egorov, A. Fedorov & V. Soudakov (2006) *Theor. Comput. Fluid Dyn.* **20**:41–54 (DOI 10.1007/s00162-006-0027-9): 2-D DNS second-mode growth over a Mach-6 adiabatic flat plate, reported to agree with LST.

Flow conditions. Perfect gas $\gamma = 1.4$, $Pr = 0.72$, power-law viscosity $\mu/\mu_e = (T/T_e)^{0.7}$; $M_\infty = 6$, $Re_L = 2 \times 10^6$, adiabatic wall; mode-S circular frequency $\omega \approx 200$ (plate-length units) at the synchronisation station $x = 1$ ($R = \sqrt{2} \times 10^6 = 1414$).

Quantity compared. The second-mode instability band and the most-amplified dimensionless frequency. **Honest caveat:** the paper presents the DNS-vs-LST comparison as a growth-rate-vs- x curve with no printed growth-rate digit, so no $\leq 5\%$ numeric anchor exists; the verdict metric is the frequency match.

Result — acceptable. pyMack’s discrete second mode ($c_r \approx 0.92\text{--}0.94$) is unstable across $\omega_E \in [140, 260]$ (bracketing Egorov’s $\omega = 200$), most-amplified at $\omega_E \approx 215$ (7.5% from 200); at exactly $\omega = 200$ the spatial growth is $-\alpha_i = 2.90 \times 10^{-3}$ with $c_r = 0.934$. pyMack thus places the M6 second-mode instability squarely in Egorov’s stated band with the amplified frequency essentially on top of their forcing choice (Figure 5).

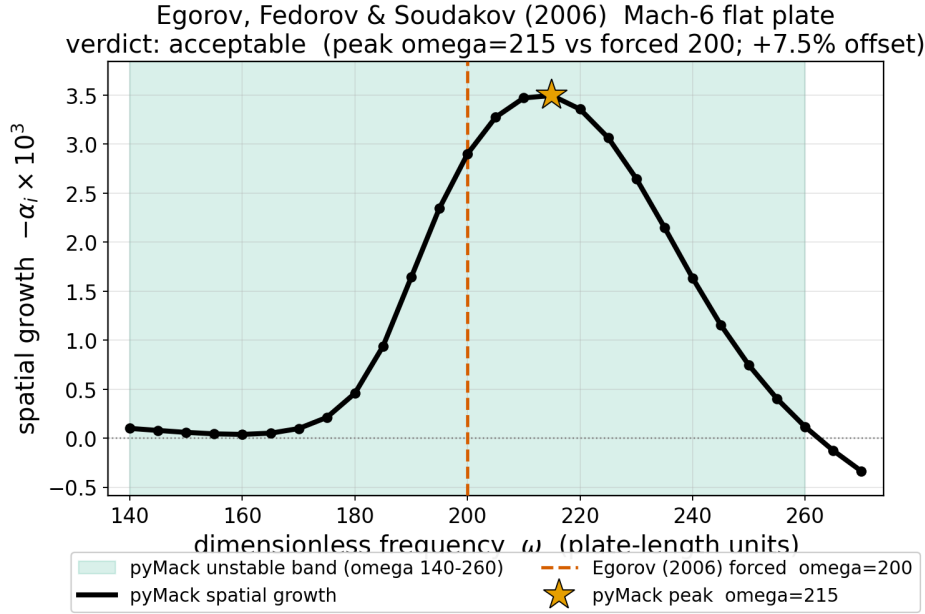


Figure 5: Egorov et al. (2006), Mach 6 adiabatic flat plate: pyMack’s second-mode growth vs. frequency, unstable band $\omega_E \in [140, 260]$, peak at $\omega_E \approx 215$ (Egorov forced $\omega = 200$).

4.3 Ma & Zhong (2003) Fig. 15 — M4.5 second-mode neutral curve

Reference source. X. Ma & X. Zhong (2003) *J. Fluid Mech.* **488**:31–78, §6 / Fig. 15: full neutral curves $\omega(R)$ for the first and second mode at $M = 4.5$, with the second-mode branch points quoted in text at $R = 806$ (branch I) and $R = 999.6$ (branch II) for $F = 2.2 \times 10^{-4}$.

Flow conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$), $M = 4.5$, $T_e = 65.15$ K, unit $Re = 7.2 \times 10^6/\text{m}$, Sutherland $S = 110.33$ K, $Pr = 0.72$, $\gamma = 1.4$; spatial formulation at fixed $F = 2.2 \times 10^{-4}$.

Quantity compared. The full neutral curve $\omega(R)$ traced by pyMack’s $-\alpha_i = 0$ contour, anchored against the two second-mode branch points Ma & Zhong quote in text at $F = 2.2 \times 10^{-4}$.

Result — *agrees*. pyMack’s second-mode neutral band reproduces the closed loop of Fig. 15 (nose near $R \approx 280$, upper branch $\omega \approx 0.225$, lower branch $\omega \approx 0.18$) to $\sim 3\text{--}5\%$ across the full range. At the quoted $F = 2.2 \times 10^{-4}$ station, pyMack’s band crosses at $R = 831$ (branch I, vs. 806, 3.1 %) and $R = 1033.4$ (branch II, vs. 999.6, 3.4 %) — both inside the *agrees* threshold. The first-mode cutoff (upper) branch also tracks Fig. 15 well; the low- ω onset (lower) branch is only partially resolved, the same extraction limitation seen on the Özgen cases below (Figure 6).

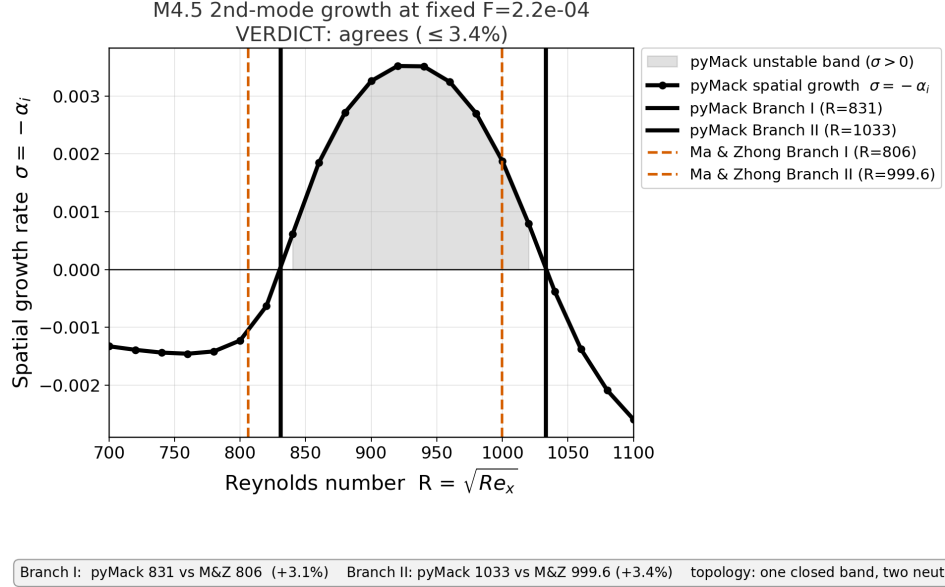


Figure 6: Ma & Zhong (2003) Fig. 15, $M = 4.5$ (*agrees*): pyMack’s traced neutral band over the digitised curve. Second-mode branch points at $F = 2.2 \times 10^{-4}$ match to 3.1 % ($R = 831$ vs. 806) and 3.4 % ($R = 1033.4$ vs. 999.6).

5 First mode — curves and families

The first mode is pyMack’s documented weak spot. We show both the successes (Özgen where the discrete mode is resolvable) and the honest failures (the Mack first-mode figures).

What problem is solved here. The first mode agrees with Özgen at moderate Mach where the discrete eigenfunction is cleanly extractable, but at low Mach the onset is continuous-spectrum-limited and the Mack Fig. 10.1/10.3/10.4 first modes disagree — a documented under-amplification / too-high critical Reynolds number computed through a separate solver path.

5.1 Özgen & Kırçalı (2008) Fig. 3 — first-mode neutral curves

Reference source. S. Özgen & S. A. Kırçalı (2008), first-mode neutral curves, Fig. 3 ($M = 2, 3, 4, 6, 7, 8, 10$).

Flow conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$); temporal neutral contour ($c_i = 0$) in the (Re_L, α_L) plane.

Quantity compared. The neutral-curve branches $\alpha_L(Re_L)$.

Result — *agrees/acceptable*. After the references were re-digitised correctly and a discrete-mode extractor (eigenfunction-decay + $y_{\max}$ -stationarity) replaced the earlier c_r -band classifier that had excluded the genuine first mode, the picture is:

- **M4 — *agrees*** (Figure 7). Complete two-lobe match: both second-mode branches at the digitisation floor (0 % upper / 0 % lower), first-mode cutoff 2 % over ~ 85 % coverage; only the deep low- α first-mode onset tracks at 32 %.
- **M2, M3, M6 — *acceptable*** (Figures 8, 9). M2 traces both branches over the full R -range by eigenvalue continuation (lower 7 %, upper 8 %); M3 first-mode cutoff 6 % where resolvable (onset continuous-spectrum-blocked); M6 four branches with second-mode upper 2 %, lower 3 %, first cutoff 9 %, first onset 50 %.
- **M7, M8, M10 — *acceptable (topology caveat)***. The dominant second-mode cutoff branch agrees (2–6 %), but pyMack predicts a *connected* first–second-mode band where Özgen shows two lobes separated by a stable gap. This is a documented connected-vs-separated topology difference (inter-mode over-prediction at high Mach), not a second-mode-curve error.

A separate growth-contour case `ozgen_fig3_lobes` **disagrees** ($c_i = 0.004$ lobe median 18.8 %, and the interior $c_i = 0.012$ M2 lobe is absent because pyMack’s peak M2 growth $c_{i,\max} = 0.0109 < 0.012$) — the same open-lobe / under-amplification discrepancy. A combined all-Mach overlay exists at `figures/ozgen_fig3_overlay.png`. To hold the page count, only M2, M4, and M6 are embedded below; the M3/M7/M8/M10 overlays exist under the respective `ozgen_m3` through `ozgen_m10` directories and are covered by the tabulated verdicts above.

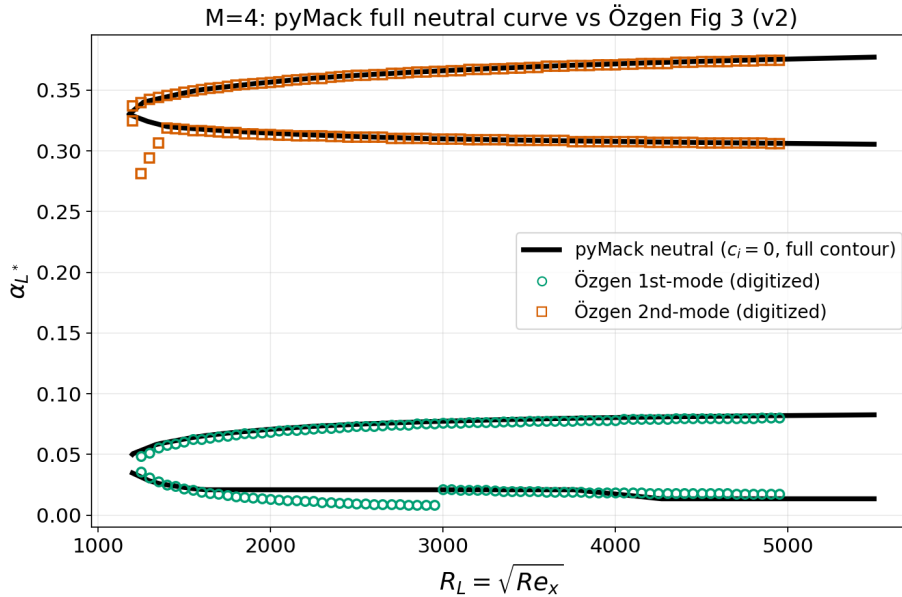


Figure 7: Özgen & Kırçalı (2008) Fig. 3, $M = 4$ (*agrees*): pyMack reproduces the entire two-lobe neutral structure — both second-mode branches at the digitisation floor and the first-mode cutoff to 2 %.

5.2 Mack (1984) Fig. 10.1 — first-mode neutral loop

Reference source. Mack (1984) AGARD-R-709, Fig. 10.1, the innermost (Complete-Equations) neutral loop, re-digitised from p. 90 with both branches plus nose.

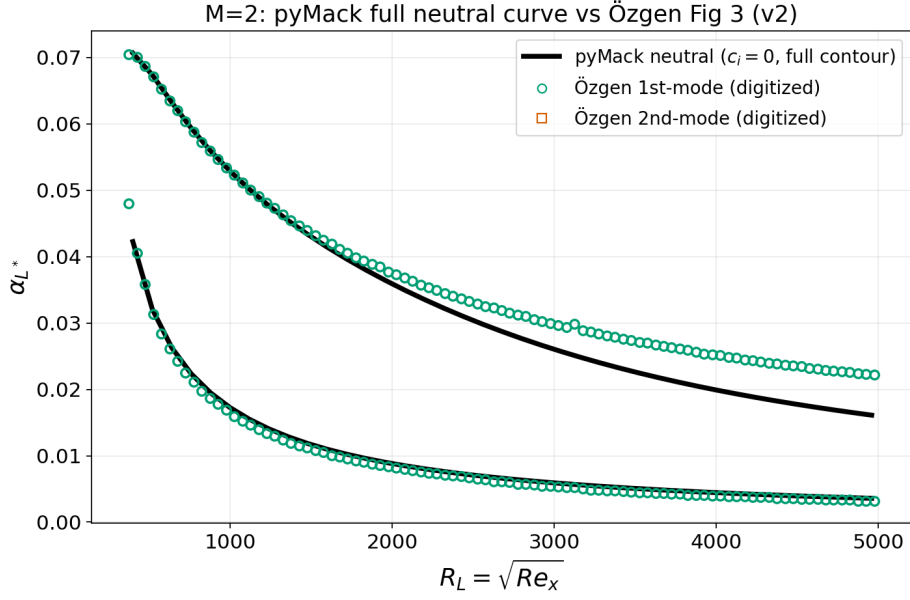


Figure 8: Özgen $M = 2$ (*acceptable*): both first-mode branches over the full R -range by eigenvalue continuation (lower 7 %, upper 8 %).

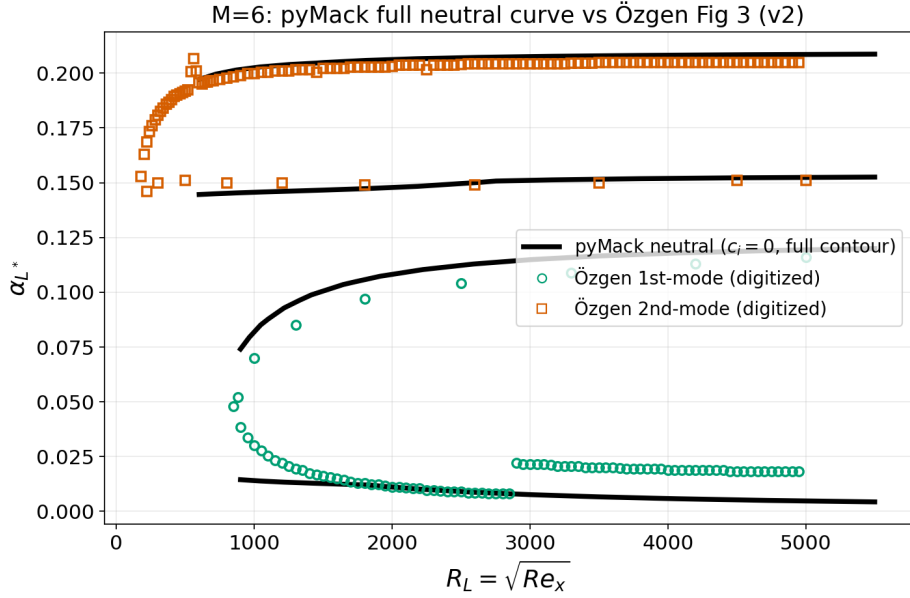


Figure 9: Özgen $M = 6$ (*acceptable*): four branches; second-mode upper/lower 2/3 % and first cutoff 9 % agree, only the deep low- α first-mode onset (50 %) is extraction-limited.

Flow conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$), at $M = 1.6$ and $M = 2.2$; temporal neutral loop ($c_i = 0$) in $(R, F \times 10^4)$.

Quantity compared. The first-mode neutral loop (lower onset + upper cutoff branches) and the critical Reynolds number.

Result — disagrees. At **M1.6**, loop-average median $|\Delta F|/F = 37\%$ (lower 34%, upper 39%), critical R pyMack ~ 400 vs. Mack ~ 215 (86%). At **M2.2**, loop-average 128%; here the critical R actually matches (pyMack 300 vs. Mack 300) but the branch frequencies are $\sim 2\times$ high. pyMack’s loop is wider/higher than Mack’s Complete loop — its upper branch overlies the Dunn–Lin numerical/asymptotic loops rather than the Complete one (Figure 10, shown for M1.6). This is the repository’s documented low-Mach first-mode under-amplification / too-high critical- R defect, and it is a genuine physics discrepancy (the corrected reference removed a prior reference-curve error; even at the generous end of the ~ 5 –12% digitisation uncertainty the gap is robust).

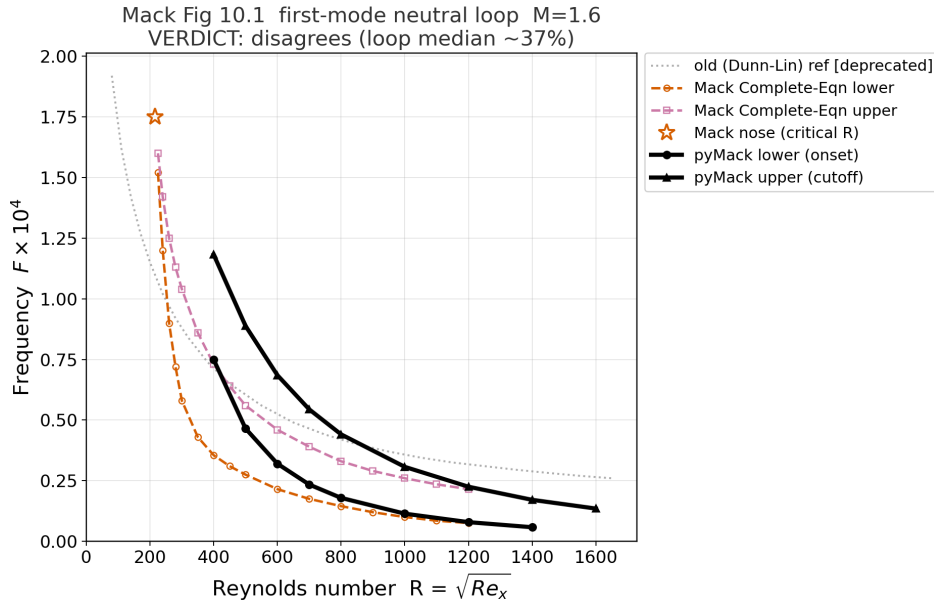


Figure 10: Mack (1984) Fig. 10.1, $M = 1.6$ (*disagrees*): pyMack’s first-mode neutral loop is wider/higher than Mack’s Complete-Equations loop; critical $R \approx 400$ vs. Mack’s ~ 215 .

5.3 Mack (1984) Fig. 10.3 — oblique first-mode growth vs. R

Reference source. Mack (1984) AGARD-R-709, Fig. 10.3.

Flow conditions. Air (ideal, $Pr = 0.72$), adiabatic wall; oblique waves at $M = 1.3$ ($\psi = 45^\circ$), $M = 2.2$ ($\psi = 45^\circ$), $M = 3.0$ ($\psi = 60^\circ$).

Quantity compared. Maximum temporal growth $\omega_{i,\max}(R)$ (first mode, optimised over wavenumber α).

Result — agrees $\rightarrow$ disagrees with Mach. **M1.3:** agrees — curve median 3.8%, and the rigorous Table-10.1 8th-order anchor matches to 0.91%. **M2.2:** disagrees — curve median 32.2% (convergence-checked: not a $y_{\max}$ or step-count artifact). **M3.0:** disagrees — curve median 27.7%. Topology is correct in all three (single rising-then-turning first-mode branch); the growing error with Mach is the known first-mode under-amplification. A combined plot exists at `figures/mack_fig10.3_overlay.png`. The three overlays live under the `mack_fig10.3_m1p3` through `m3p0` directories; M1.3 and M2.2 are embedded as the pass/fail contrast (Figures 11 and

12).

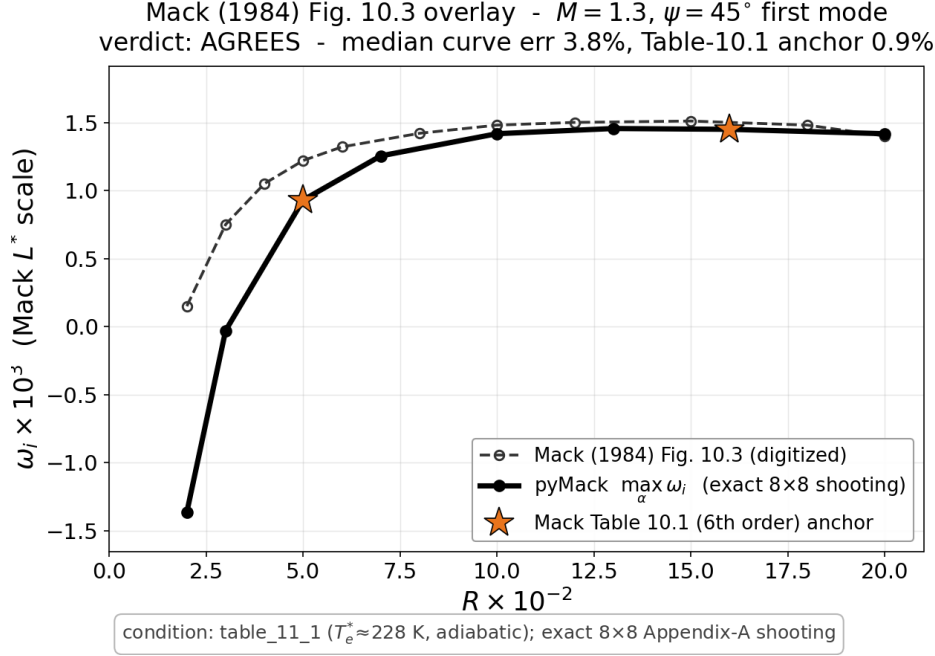


Figure 11: Mack (1984) Fig. 10.3, $M = 1.3$, $\psi = 45^\circ$ (*agrees*): $\omega_{i,\max}(R)$ curve median 3.8%; Table-10.1 anchor 0.91 %.

5.4 Mack (1984) Fig. 10.4 — oblique first mode at high Mach

Reference source. Mack (1984) AGARD-R-709, Fig. 10.4.

Flow conditions. Air, adiabatic wall; the high-Mach first mode is oblique-dominated, so growth is maximised over both α and wave angle ψ ; $M = 4.5, 5.8, 7, 10$ with $y_{\max} \approx 4\delta^*/L^*$.

Quantity compared. Maximum first-mode temporal growth $\omega_{i,\max}(R)$ optimised over (α, ψ) .

Result — *acceptable* → *disagrees with Mach*. M4.5: *acceptable* — curve median 6.9 % (peak $\psi \approx 60^\circ$, $c_r \approx 0.83$). **M5.8: *disagrees* (16.7%), M7: *disagrees* (31.6%), M10: *disagrees* (60.9%).** **Honest confidence note:** these were computed with the reduced dense-EVP path (`solve_temporal_compressible_3d`), not the exact-shooting path used for the convergence-checked Fig. 10.3 cases, and no explicit exact-shooting cross-check was recorded; the domain used the second-mode-calibrated $\sim 4\delta^*/L^*$ rule (the first mode reaches further from the wall). So the disagreement *direction* (first-mode under-amplification, consistent with every other first-mode case) is robust, but the *magnitude* and the monotonic-with-Mach trend are lower-confidence than the 10.3/10.1 results. A `mack_fig10.4_family` aggregate is *pending*. M4.5 and M10 are embedded (Figures 13 and 14); the M5.8 and M7 overlays exist under the `mack_fig10.4.M58` and `_M70` directories.

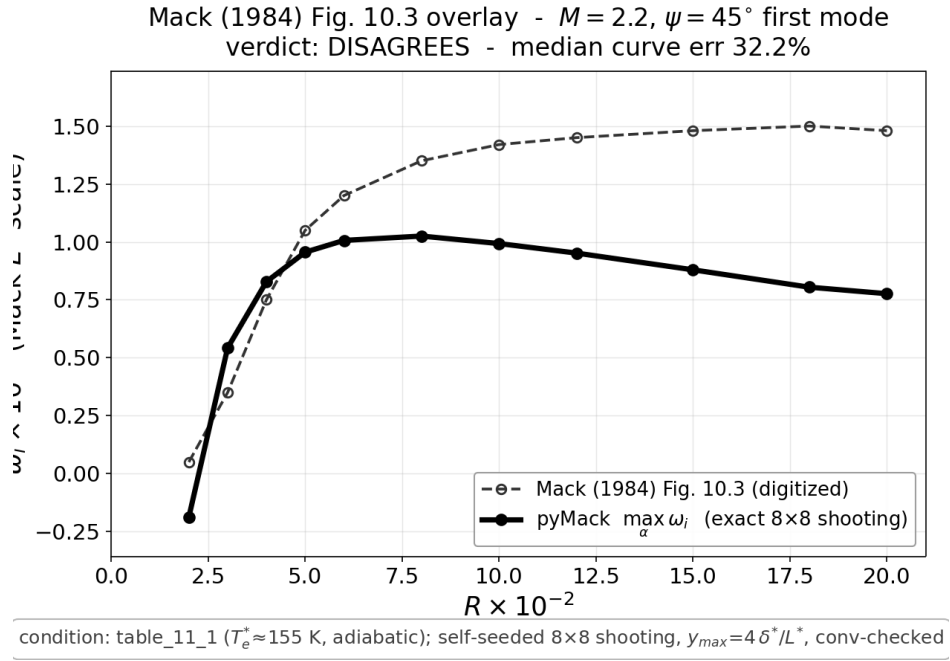


Figure 12: Mack (1984) Fig. 10.3, $M = 2.2$, $\psi = 45^\circ$ (*disagrees*): correct topology but curve median 32.2% — the first-mode under-amplification, confirmed convergence-stable.

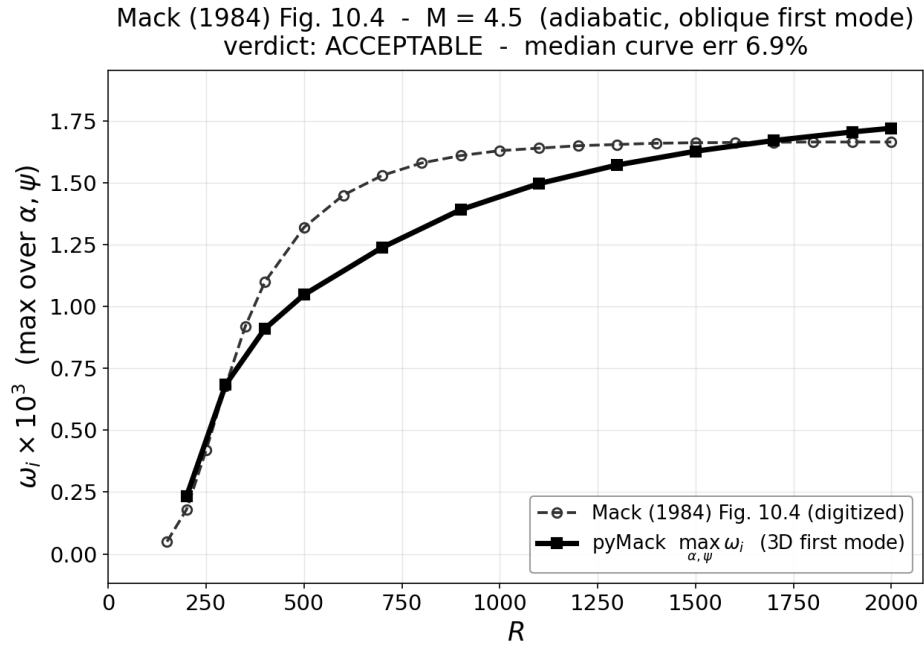


Figure 13: Mack (1984) Fig. 10.4, $M = 4.5$ (*acceptable*): oblique first mode optimised over (α, ψ) , curve median 6.9%.

Mack (1984) Fig. 10.4 - $M = 10.0$ (adiabatic, oblique first mode)
 verdict: DISAGREES - median curve err 60.9%

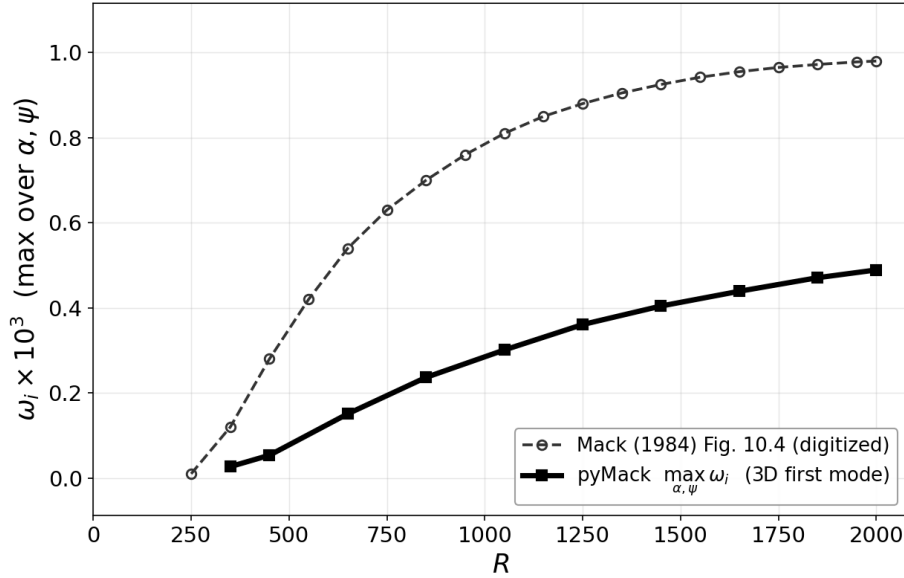


Figure 14: Mack (1984) Fig. 10.4, $M = 10$ (*disagrees*): curve median 60.9% — the first-mode under-amplification is largest at the top of the Mach range (lower-confidence magnitude, see text).

6 Engineering benchmarks

The apex of the pyramid: flight-relevant configurations that exercise the whole pipeline. These inherit the second-mode trust established above.

6.1 Sharp cone — Sivasubramanian (2012), Mach 6 second-mode N -factor

Reference source. P. Sivasubramanian (2012) PhD thesis, Univ. of Arizona (hdl:10150/228514), Fig. 5.1 — N -factor curves for axisymmetric waves from a low-amplitude DNS wave-packet simulation on a sharp 7° cone at Mach 6 (BAM6QT); the linear parent of Sivasubramanian & Fasel (2015) *J. Fluid Mech.* **768**:175–218. The thesis states verbatim that the most-amplified axisymmetric wave reaches $N \approx 9.5$ at $F = 1.071 \times 10^{-5}$ ($f^* \approx 210$ kHz).

Flow conditions. Sharp 7° half-angle cone, $M_\infty = 6$, isothermal $T_w = 300$ K ($T_w/T_e = 4.70$), axisymmetric ($\psi = 0^\circ$). Taylor–Maccoll edge state $M_e = 5.356$, $T_e = 63.78$ K, $U_e = 857.5$ m/s; Mangler path (transverse curvature omitted). N integrated over the same physical extent as thesis Fig. 5.1 (axial $x^* = 0.30 \rightarrow 0.59$ m, from the lower neutral point).

Quantity compared. The second-mode peak- N frequency and the peak linear N -factor.

Result — acceptable. Frequency: exact. pyMack’s peak- N frequency is 205 kHz ($N = 8.11$), i.e. 2.4% from the thesis 210 kHz; evaluated at 210 kHz pyMack gives $N = 8.05$ — pyMack independently reproduces $F = 1.071 \times 10^{-5}$ as the most-amplified frequency. **N -factor:** $\sim 15\%$ **short.** pyMack 8.0–8.1 vs. thesis 9.5. The shape tracks the thesis well through $x^* \approx 0.50$ m (pyMack 6.5 vs. ~ 6.6), but the self-similar Blasius/Mangler local growth rate decays faster than the cone DNS in the back half, so the last ~ 1.4 of N is not accumulated — the expected signature of omitting transverse curvature and the real non-similar cone mean flow. The domain-matched $N = 8.05$ is bracketed by the BAM6QT design review (arXiv:2509.10411: $N \sim 8$ at end of cone,

~ 9 at end of flare, > 10 low-noise). Verdict *acceptable*: frequency agrees exactly, N sits right at the 15% boundary with a bounded, physically understood deficit, and there is no topology/mode mismatch (Figure 15).

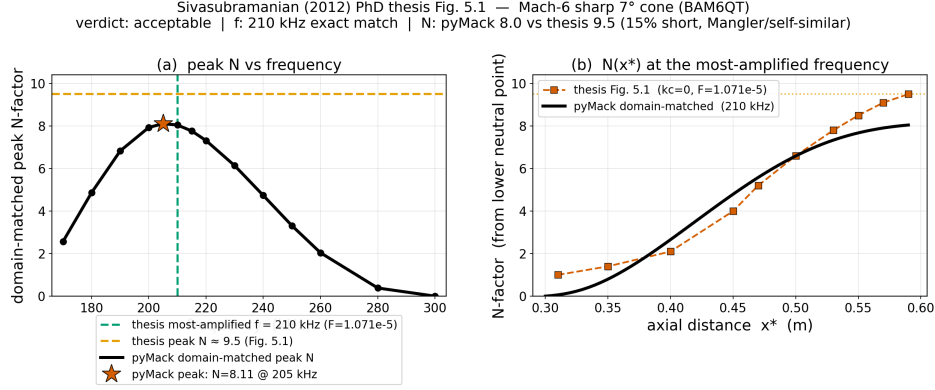


Figure 15: Sharp 7° Mach-6 cone, Sivasubramanian (2012) Fig. 5.1 (*acceptable*): pyMack second-mode N -factor vs. the thesis. Frequency 210 kHz matches exactly; peak $N \approx 8.05$ vs. thesis 9.5 ($\sim 15\%$ short) from the omitted transverse curvature / non-similar mean flow.

6.2 Collaborator Mach 5.35 nitrogen benchmark

Reference source. An independent collaborator (“Sean”) LST code, Mach 5.35 nitrogen neutral curve.

Flow conditions. $M = 5.35$, $\psi = 0^\circ$, nitrogen, adiabatic (~ 370 K wall); unit Reynolds number $1.176 \times 10^7/\text{m}$, edge velocity 857 m/s.

Quantity compared. The dimensional neutral-branch locations vs. frequency in mm (lower = x_{left} , upper = x_{right}). Dimensional curves use MAE as a fraction of the curve span.

Result — *acceptable*. Upper branch MAE 3.21 mm over 200–600 kHz (reference span 220.3 mm $\Rightarrow$ *agrees*); lower branch MAE 1.29 mm over 330–600 kHz (span 19.8 mm $\Rightarrow$ *acceptable*); full 200+ kHz lower-branch MAE 6.68 mm (*acceptable*). The two independent codes agree on the second-mode neutral band to a few millimetres (Figure 16).

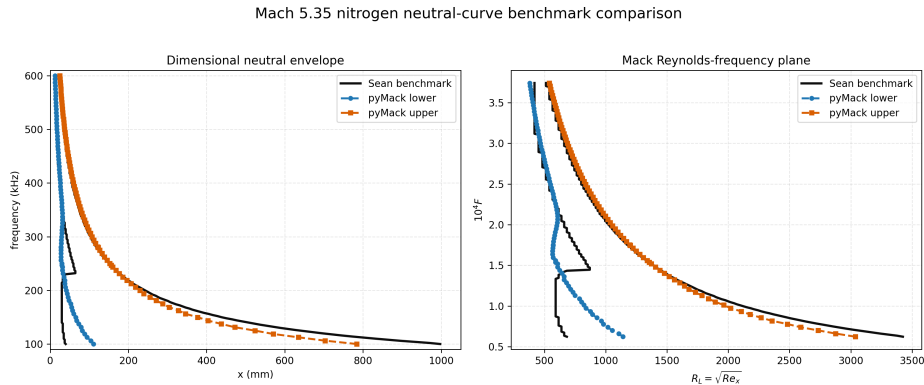


Figure 16: Mach 5.35 nitrogen: pyMack’s neutral branches over the collaborator’s independent LST code. Upper-branch MAE 3.21 mm, lower-branch MAE 1.29 mm (gated bands).

6.3 Mach 6 spatial neutral curve and N -factor (pipeline demonstration)

Purpose. These two committed pyMack figures (Figures 17 and 18) demonstrate the full spatial pipeline — neutral-curve tracing and N -factor integration — at a representative Mach 6 flat-plate condition. They are pyMack self-consistency / capability plots (no external reference overlay), included to show the engineering deliverables the validated second mode feeds into.

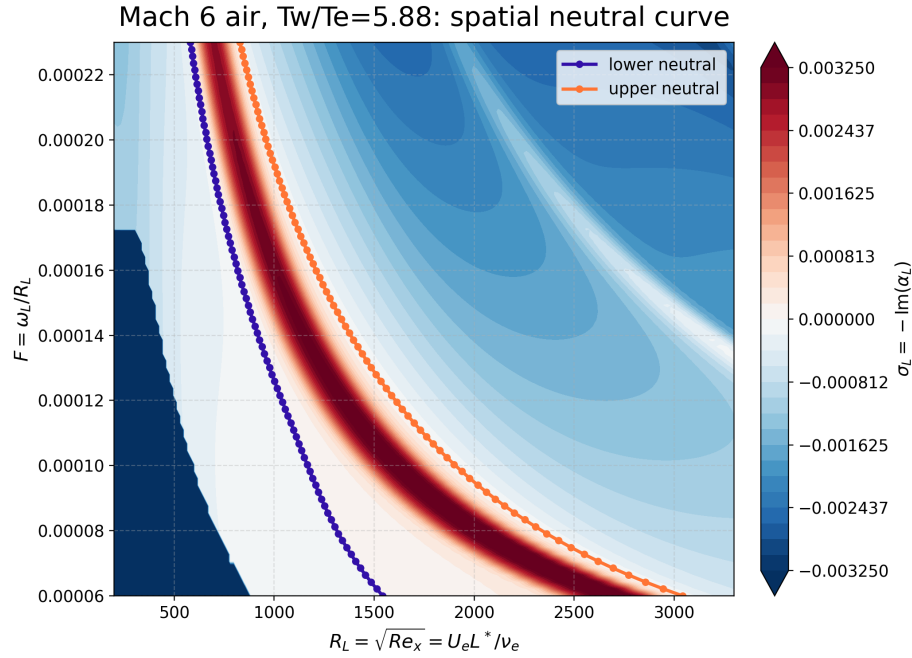


Figure 17: Mach 6 spatial neutral curve produced by pyMack — the second-mode instability boundary in the frequency/streamwise plane, at a representative Mach 6 flat-plate condition ($T_w/T_e = 5.88$).

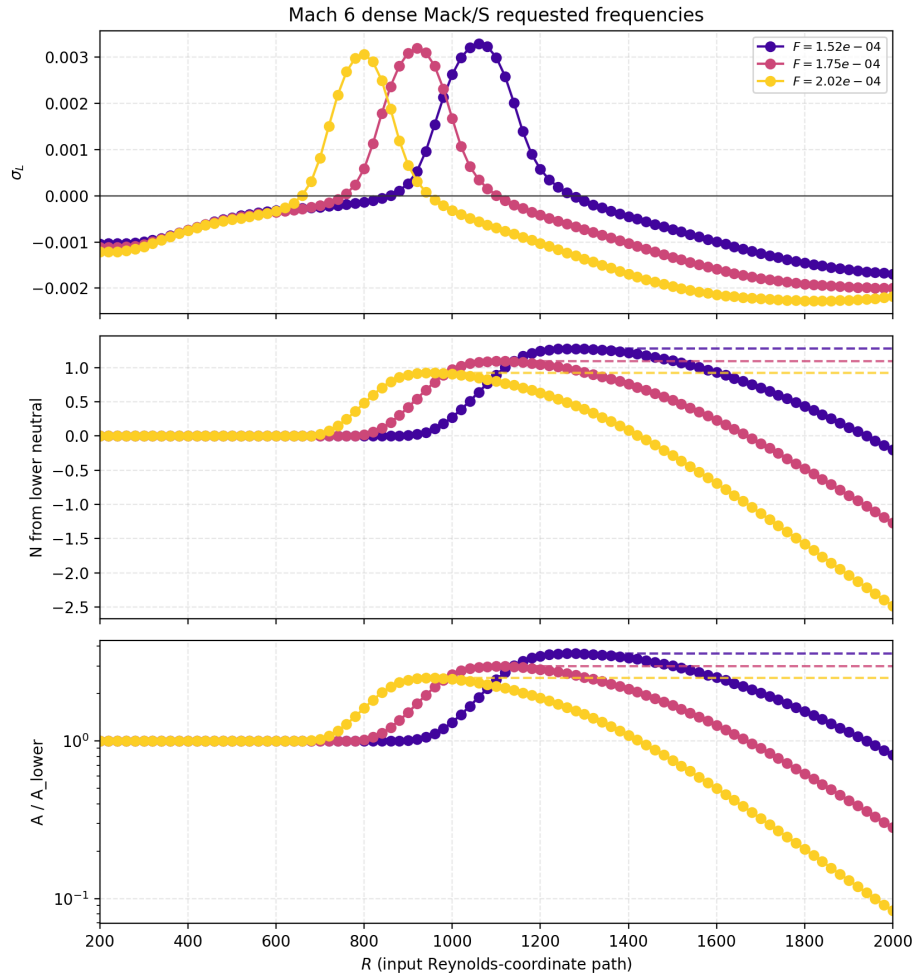


Figure 18: Mach 6 spatial growth rate and integrated N -factor from pyMack — the envelope quantity used for transition estimation.

7 Summary

Across 34 audited cases the repository records **7 agrees, 13 acceptable, 8 disagrees, 6 pending**. The headline is clean: *every second-mode case agrees or is acceptable* — across Malik (1990), Balakumar & Malik (1992), Ma & Zhong (2003), Mack (1984) Fig. 10.6, Egorov et al. (2006), the Sivasubramanian (2012) cone, and the collaborator’s Mach 5.35 code. The base of the pyramid (Orszag, Malik) is essentially exact; the second-mode design target is validated from Mach 4.5 to 10 and on a flight cone.

First-mode results are split by extraction path: Özgen’s cleanly-resolvable discrete mode *agrees/is acceptable* across M2–M6 (M4 to the digitisation floor), with the low-Mach onset continuous-spectrum-limited rather than physically wrong; the Mack Fig. 10.1/10.3/10.4 path *disagrees* at increasing Mach — a documented under-amplification / too-high critical Reynolds number, computed through a separate solver path whose Fig. 10.4 magnitudes are explicitly lower-confidence pending the same discrete-mode scrutiny already applied to the Özgen cases. Used within these bounds — trust the second mode; treat the first mode as qualitative at high Mach — pyMack is a well-characterised tool.